

Jatin Chowdhury

Audio Signal Processing Engineer

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Education

Massachusetts Institute of Technology

Ph.D in Electrical Engineering and Computer Science

Cambridge, MA

Sept. 2025 - PRESENT

- Relevant research: real-time circuit simulation, audio software development, signal processing.

Stanford University, Center for Computer Research in Music and Acoustics

M.A. in Music, Science, and Technology

Palo Alto, CA

Sept. 2018 - June 2020

- Relevant coursework: physical audio signal processing, time-frequency audio signal processing, machine learning, digital instrument design, spatial audio, music recording.
- Relevant research: physical modelling, nonlinear filter design, modal modelling, machine learning.
- Dunning Family Fellowship for the 2018-2019 academic year.

University of Southern California

B.S. in Electrical Engineering

Los Angeles, CA

Aug. 2014 - May 2018

- Completed Minors in Physics and Music Recording.
- Relevant coursework: signal processing, circuit design, digital logic, electromagnetics, software design.
- USC Presidential Scholarship for outstanding academic achievement.
- USC Renaissance Scholar certificate for excelling academically while pursuing separate fields of study.

Experience

Chowdhury DSP

Audio Signal Processing Engineer

Renton, WA

Oct. 2021 - PRESENT

- Developed signal processing algorithms for real-time audio processing and synthesis.
- DSP algorithms include: virtual analog modelling, machine learning, filter design, and frequency-domain processing.
- Developed audio plugins including DSP, UI implementation, preset management, copy protection, installers, etc.
- Rendered services for clients including: BABY Audio, Schwabe Digital, Chandler Limited, Bogren Digital, Little Sir Rekko, VCV, KIT Plugins, Sabol Hill, Submission Audio, ADPTR Audio, and more.
- Hired, mentored, and trained interns in audio signal processing and software development.

Tesla Motors

Audio Test Engineer

Palo Alto, CA

Aug. 2020 - Oct. 2021

- Developed audio system signal flow layouts for vehicles using AudioWeaver.
- Contributed to audio system testing and tuning including telephony, equalization, and spatialization.
- Developed end-of-line tests to ensure vehicle audio system quality using Python and C++.
- Contributed to the signal processing and tuning for the vehicle pedestrian warning speaker.
- Contributed to testing and validation software for audio system firmware.

Pers3ctive VR

Software/DSP Engineer

Los Angeles, CA

June 2019 - Mar. 2020

- Developed and implemented audio effects for use in a VR audio production environment.
- Implemented the audio effects as an audio plugin with a designed GUI.
- Built DSP framework for integrating effects made by other engineers into the plugin.
- Implemented OSC communication for sharing parameters and metering data with VR headset.

Audioworks Technologies

Software Engineer

Toronto, ON

July 2018 - Dec. 2018

- Member of of the SoundsUnite developement team: building a digital audioworkstation (DAW) using JUCE/C++.
- Developed DSP features for the application including panning algorithms, level detection, and "smart" track exporting.
- Developed UX features for the application including meters, file browsers, and tempo controls.
- Integrating the application file management system with the SoundsUnite web store.

McGill Space Institute

Research Trainee

Montreal, QC

May 2017 - Aug. 2017

- Member of the CHIME/FRB Working Group: building a software pipeline to detect Fast Radio Bursts (FRBs).
- Developed a "Flux Estimator" module for the pipeline to estimate the intrinsic brightness of astrophysical events.
- Contributed to science and unit testing frameworks for the CHIME/FRB software pipeline.
- Presented work for the CHIME/FRB Working Group and CHIME/FRB Pipeline Gamma Release.

Skills

Programming Languages

C/C++, Python, Bash, Jai, MATLAB, Faust, LaTeX

Signal Processing

Virtual analog modelling, filter design, machine learning, time-frequency analysis

Frameworks

JUCE, CLAP, TensorFlow, PyTorch, Arduino

Tools

CMake, Git, Linux CLI, Visual Studio, Xcode, GitHub Actions, Inno Setup

Publications

Performance-Oriented Wave Digital Circuit Emulation

Proc. of the 29th International Conference on Digital Audio Effects

Sept. 2026
Cambridge, USA

- Co-authored with Mark Rau.

Available on the DAFx Archives

Inference-Time Structured Pruning for Real-Time Neural Network Audio Effects

Proc. of the 28th International Conference on Digital Audio Effects

Sept. 2025

Ancona, IT

- Co-authored with Christopher Johann Clarke.

Available on the DAFx Archives

Sample Rate Independent Recurrent Neural Networks for Audio Effects Processing

Proc. of the 27th International Conference on Digital Audio Effects

Sept. 2024

Guildford, UK

- Co-authored with Alistair Carson, Alec, Wright, Vesa Välimäki, and Stefan Bilbao.

Available on the DAFx Archives

Computationally Efficient Physics Approximating Neural Networks for Highly Nonlinear Maps

Proc. of the Conference on Research in Adaptive and Convergent System

Oct. 2022

New York, USA

- Co-authored with Christopher Johann Clarke and others.

Available on the ACM Digital Library

chowdsp_wdl: An Advanced C++ Library for Wave Digital Circuit Modelling

ArXiv E-Prints: Audio and Speech Processing

Oct. 2022

Online

- Presents a wave digital circuit modelling library with fixed-topology optimizations and other advancements.

Available on the ArXiv

Emulating Diode Circuits with Differentiable Wave Digital Filters

19th Sound and Music Computing Conference

June 2022

Saint Etienne, France

- Co-authored with Christopher Johann Clarke.

Available on Zenodo

A Wave Digital Filter Modeling Library for the Faust Programming Language

Proc. of the 18th Sound and Music Computing Conference

June 2021

Torino, IT

- Co-authored with Dirk Roosenburg, Eli Stine, and Romain Michon.

Available on GitHub

RTNeural: Fast Neural Inferencing for Real-Time Systems

ArXiv E-Prints: Audio and Speech Processing

June 2021

Online

- Presents a neural network inferencing library for real-time systems.

Available on the ArXiv

Water Bottle Synthesis with Modal Signal Processing

Proc. of the 23rd International Conference on Digital Audio Effects

Sept. 2020

Vienna, Austria

- Presented at the DAFx-2020 conference.

Available on the DAFx Archives

Stable Structures for Nonlinear Biquad Filters

Proc. of the 23rd International Conference on Digital Audio Effects

Sept. 2020

Vienna, Austria

- Presented at the DAFx-2020 conference.

Available on the DAFx Archives

Real-Time Physical Modelling for Analog Tape Machines

Proc. of the 22nd International Conference on Digital Audio Effects

Sept. 2019

Birmingham, UK

- Presented at the DAFx-2019 conference.

Available on the DAFx Archives

The CHIME Fast Radio Burst Project: System Overview

The Astrophysical Journal

Aug. 2018

Vol. 836, No. 1

- Co-authored with the CHIME/FRB Collaboration

- Available on the ArXiv.

Professional & Academic Service

Peer Reviewer

Provided peer review for papers submitted to conferences and journals, including DAFx, Audio Mostly, and the EURASIP Journal on Audio, Speech, and Music Processing.

Organizer

Served on the local organizing committee for the 29th International Conference on Digital Audio Effects (DAFx26), in Cambridge, MA.

Mentor

Participated in the Audio Developer Conference Mentorship Program in 2025 and 2026.

Presentations

Wait-Free Programming From Scratch

Northwest C++ User Group Meetup

June 2026

Bellevue, WA

- Discussed the challenges of designing wait-free algorithms for audio and other real-time software applications.
- Discussed the trade-offs of well-known solutions including spin mutexes and lock-free queues.
- Presented the development of a "from scratch" wait-free solution to a specific problem.

AI Inference for Real-Time Audio Applications

Boston AI Music Meetup

February 2026

Cambridge, MA

- Discussed the challenges of developing real-time systems in the context of music software.
- Discussed the challenges of performing neural network inference within a real-time system.

Performance Engineering for Modern DSP

Signal Processing Summit

September 2025

Milpitas, CA

- Discussed performance engineering for signal processing engineers and software developers.
- Presented recommendations based on modern advancements in CPU and DSP hardware technology.
- Discussed ideas for teaching novice engineers about performance.

Real-Time Audio Programming with Memory Arenas

The Audio Programmer Monthly Meetup

October 2024

Online

- Presented a brief overview of heap and stack memory allocation.
- Introduced the concept of the “arena allocator” and some of its properties.
- Discussed the use of arena allocators for real-time audio programming applications.

Productive Programming with the JAI Programming Language

Northwest C++ User Group Meetup

October 2024

Redmond, WA, USA

- Presented a brief introduction to the JAI programming language.
- Discussed relevant JAI features, including compile-time meta-programming and the JAI `Context`.
- Discussed the use of JAI in audio programming applications.

Real-Time Circuit Simulation with Wave Digital Filters in C++ (Poster)

CppCon

September 2024

Aurora, CO, USA

- Discussed the basic formulation of Wave Digital Filters (WDFs) as a technique for modelling analog circuits.
- Described the usage of C++ compile-time meta-programming for implementing WDFs.
- Compared the runtime performance of various WDFs written in C++ and other languages.
- Presented a simple synthesizer made entirely with WDFs.

A Comparison of Virtual Analog Modelling Techniques (Part 2)

Audio Developer Conference

November 2023

London, UK

- Co-presented with Christopher Johann Clarke.
- Presented a virtual analog model of the Tube Screamer distortion circuit using multiple modelling techniques.
- Discussed white-box methods: Modified Nodal Analysis, Nodal DK, and Wave Digital Filters with R-Type adaptors.
- Discussed black-box methods: Recurrent Neural Networks, conditional models, and transformers.
- Discussed grey-box methods: Symbolic Regression, Differentiable WDFs, and Physics-Approximating Neural Nets.

Audio Buffers, Buffer Views, and Buffer Iterators

The Audio Programmer Monthly Meetup

September 2023

Online

- Reviewed data structure design in C++, focusing on owning and non-owning containers.
- Discussed the design of an “owning” audio buffer data structure, and a “non-owning” buffer view data structure.
- Discussed the implementation of custom C++ iterators for audio buffer data structures.

Real-Time Circuit Simulation with Wave Digital Filters

Northwest C++ Users Group Meetup

July 2023

Redmond, WA

- Discussed the challenges of real-time audio programming.
- Discussed the basic formulation of Wave Digital Filters (WDFs) as a technique for modelling analog circuits.
- Described the usage of C++ run-time polymorphism and compile-time meta-programming for implementing WDFs.
- Demonstrated example applications of WDF models in developing digital audio effects.

Replacing the AudioProcessorValueTreeState

The Audio Programmer Monthly Meetup

December 2022

Online

- Discussed the historical and current methods of audio plugin state management with the JUCE framework.
- Proposed a new approach for managing plugin state with an emphasis on real-time safety.

Physical Modelling for Analog Tape Emulation

Acoustical Society of America Annual Meeting

December 2022

Nashville, USA

- Discussed the development of a physical-modelling algorithm for real-time tape emulation.
- Discussed the physical basis for modelling magnetic hysteresis with nonlinear differential equations.
- Discussed implementation challenges for real-time tape emulation algorithms.

Analog Modelling with Wave Digital Filters in C++

The Audio Programmer Monthly Meetup

December 2021

London, UK

- Discussed the basic formulation of Wave Digital Filters (WDFs) as a technique for modelling analog circuits.
- Discussed two approaches for implementing WDFs in C++, with trade-offs between flexibility and performance.
- Demonstrated example applications of WDF models in developing digital audio effects.

CI/CD for Audio Plugin Development

Audio Developer Conference

November 2021

London, UK

- Discussed the basic concepts and motivations for continuous integration and continuous deployment (CI/CD).
- Discussed the usefulness of CI/CD for developing audio plugins.

Demonstrated several CI/CD workflows in the context of audio plugin development.

Creating Audio Effect Plugins for the 21st Century

SARC Virtual Events Series

February 2021

Belfast, UK

- Presented at the Sonic Arts Research Centre (SARC) at Queen’s University Belfast.
- Presented a paradigm for developing original and innovative digital audio software.
- Discussed techniques and philosophies for designing audio software as part of a creative process, including signal processing algorithms and user interfaces.

A Comparison of Virtual Analog Modelling Techniques

Audio Developer Conference

November 2020

London, UK

- Presented a virtual analog model of the Klon Centaur guitar pedal developed using multiple modelling techniques, including nodal analysis, Wave Digital Filters, and Recurrent Neural Networks.

An accompanying paper is published on the ArXiv.

Complex Nonlinearities for Audio Signal Processing

CCRMA DSP Seminar

May 2019

Stanford, CA

- Presented at the Center for Computer Research in Music and Acoustics (CCRMA) at Stanford University.
- Presented a series of audio effects constructed from innovative uses of nonlinear signal processing techniques.

Exploring Real-Time DSP Systems for Mixing and Performance

CCRMA DSP Seminar

Apr. 2019

Stanford, CA

- Presented at the Center for Computer Research in Music and Acoustics (CCRMA) at Stanford University.
- Presented overview of previous research and projects to an audience of peers, professors, and the public.
- Discussed topics including real-time timbral conversion, methods of integrating “imperfection” into audio DSP systems, and designing user interfaces for DSP systems as an extension of the DSP systems themselves.

Research

Code Generation Workflows for Circuit Emulation

Lead Researcher

Cambridge, MA

June 2025 - PRESENT

- Developed code generation systems for generating optimized implementations of circuit models using either Wave Digital Filters or nodal analysis.
- Developed a textual representation of Wave Digital circuit topology.
- Developed analysis tools for “reducing” circuit model complexity.

Real-Time Neural Network Inferencing

Lead Researcher

Stanford, CA

June 2020 - PRESENT

- Developed RTNeural, a C++ library for performing neural inferencing for real-time audio systems.
- Developed performance benchmarks to compare the library with LibTorch, ONNX Runtime, and TensorFlow Lite.

Differentiable Wave Digital Filters

Lead Researcher

July 2021 - Sept. 2022

- Demonstrated a process for constructing Wave Digital Filter (WDF) circuit models, in conjunction with neural network models of specific circuit elements.
- Developed a Python library for creating differentiable WDF models, allowing them to be trained via backpropagation.

Project documentation is available on GitHub.

Klon Centaur Emulation

Lead Researcher

Stanford, CA

Mar. 2020 - Aug. 2020

- Developed a digital emulation of the Klon Centaur guitar distortion pedal.
- Used circuit modelling methods including nodal analysis, Wave Digital Filters, and Recurrent Neural Networks.
- Developed emulation as an audio plugin and guitar pedal using an Arduino Teensy microcontroller.

Project documentation is available on GitHub.

Modal Waterbottles

Researcher

Stanford, CA

June 2019 - April 2020

- Worked with a team of researchers to develop acoustical models of water bottles using modal signal processing.
- Developed a real-time water bottle synthesizer using the JUCE framework.

Project documentation is available on GitHub.

Complex Nonlinearities

Lead Researcher

Aug. 2019 - June 2020

- Built frameworks for developing complex nonlinear audio signal processors.
- Developed a series of open-source plugins to demonstrate the nonlinear frameworks.
- Published a series of Medium articles explaining nonlinear signal processing techniques for a non-technical audience.

Project documentation can be found on GitHub.

Bad Circuit Modelling

Lead Researcher

Nov. 2019 - June 2020

- Researched methods for virtual analog circuit modelling to account for imperfections that exist in real-world circuits.
- Developed simulations and real-time circuit-modelling systems that accurately model the aging and component tolerances of resistors and capacitors.

Project documentation can be found on GitHub.

Analog Tape Modelling

Lead Researcher

Stanford, CA

Jan. 2019 - Apr. 2019

- Researched and developed a physical model of an analog, reel-to-reel tape machine.
- Modelled the nonlinear process of analog tape recording using the Jiles-Atherton model for magnetic hysteresis.
- Modelled other effects created by analog tape machines, including flutter, playhead loss effects, and biasing effects.
- Developed an open-source audio plugin implementing the analog tape model as a real-time system. Source code and documentation are available on GitHub.

notGuitar

Research Partner

Los Angeles, CA

Jan. 2018 - May 2018

- notGuitar is real-time timbral conversion system designed to process a guitar input signal to sound like a saxophone.
- notGuitar was implemented using a Texas Instruments DSK6713 DSP board in May 2018.
- Documentation for the project can be found on GitHub.

Projects

Chowdhury DSP

Founder, Engineer

June 2018 - PRESENT

- Developed open-source audio plugins including:
 - ChowTapeModel: An analog tape emulation based on the DAFx-19 paper “Real-Time Physical Modelling for Analog Tape Machines”.
 - BYOD: A modular system for creating guitar effect signal chains.
 - ChowMatrix: A dynamic matrix of stereo delay effects.
 - ChowKick: A kick drum synthesizer based on creative circuit modelling.
 - ChowCentaur: A digital emulation of the Klon Centaur distortion pedal, using Wave Digital Filters and recurrent neural networks.
 - ChowPhaser: A phaser effect based loosely on the Schulte Compact Phasing ‘A’ circuit.
- Developed effects for guitar pedal platforms including the Aviate Audio Multiverse and Chaos Audio Stratus pedals.
- Contributed audio effects and other DSP code to the open-source Surge Synthesizer project.
- Developed modules for the VCV Rack modular synthesis environment.

For more information, see the Chowdhury DSP website.

CLever Audio Plugins (CLAP)

Contributor

Mar. 2022 - PRESENT

- Contributed to the CLAP plugin standard with testing and documentation.
- Proposed new “extensions” to the CLAP standard.
- Contributed to a wrapper to allow developers using the JUCE framework to export CLAP plugins.
- Contributed to a JUCE module allowing JUCE applications to host CLAP plugins.

Wave Digital Filters

Programmer

Jan. 2020 - PRESENT

- Created a minimal C/C++ Library for modelling circuits with Wave Digital Filters (WDFs).
- Currently includes WDF implementations for basic circuit elements (resistors, capacitors, inductors), basic circuit adaptors (series, parallel), “1-port” nonlinear circuit elements, and “R-Type” adaptors.
- Includes several example WDF models implemented as audio plugins using the JUCE framework, as well as a graphical WDF prototyping tool.
- Developed an auxiliary script for deriving scattering matrices for “R-Type” adaptors.

Ongoing development and documentation can be found on GitHub.

NoLava Recording Studios

Audio Engineer

Los Angeles, CA

Aug. 2017 - July 2018

- Recorded, mixed, and mastered for artists of various styles including acoustic, electronic, rock, country, and punk.
- Installed, repaired, and maintained speakers, microphones, keyboards, amplifiers, and other studio equipment.